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by

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Autumn Semester, 2026–27

November 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

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DEPARTMENT OF INDUSTRIAL AND SYSTEMS ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled "**Risk-Aware Deep Reinforcement Learning for Daily Portfolio Allocation**" submitted by **Ritam Mondal** (Roll No. 23IM30018) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Industrial and Systems Engineering is a record of bona fide work carried out under my supervision and guidance during Autumn Semester, 2026–27.

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Acknowledgement and Abstract

Acknowledgement:

I express my deepest gratitude to my supervisor, Professor Abhisek Sharma, for his insightful guidance, intellectual mentorship, and continuous encouragement throughout this B.Tech. Project. His domain expertise in financial engineering, operations research, and sequential decision processes has been instrumental in shaping the theoretical and empirical methodology of this thesis. I also extend my gratitude to the Department of Industrial and Systems Engineering, Indian Institute of Technology Kharagpur, for providing the necessary computing resources, and to my peers and family for their unwavering support.

Abstract:

Daily portfolio allocation is difficult to automate because financial markets exhibit low signal-to-noise ratios, non-stationary dynamics, and only partial observability of the true underlying economic state. Deep reinforcement learning (DRL) research has separately proposed temporal memory, risk-aware reward shaping, and turnover regularization to address these challenges, but such mechanisms are rarely isolated under identical experimental conditions, making it difficult to attribute performance differences to any single design choice.

This report presents a rigorous, controlled four-stage ablation framework: (i) Model M1: Proximal Policy Optimisation (PPO) with a feedforward Multi-Layer Perceptron (MLP) baseline and net return reward; (ii) Model M2: PPO with a recurrent Long Short-Term Memory (LSTM) encoder and net return reward; (iii) Model M3: LSTM-PPO with recursive Differential Sharpe Ratio (DSR) reward shaping; and (iv) Model M4: LSTM-PPO with DSR reward and explicit quadratic turnover regularisation ($\lambda_{\text{turn}} = 0.1$). All models parameterise portfolio allocations over a continuous Dirichlet distribution, naturally satisfying non-negativity and budget simplex constraints with an explicit cash reserve.

Models are trained and evaluated on the same daily equity universe of ten major National Stock Exchange of India (NSE) stocks, identical thirteen causal technical indicators, realistic proportional transaction costs (0.1%), and four rolling walk-forward folds across 2018–2025 over multiple random seeds. Results reveal that temporal memory (M2) substantially enhances out-of-sample Sharpe ratio (+0.201) while cutting daily turnover by 57%. Turnover regularisation (M4) effectively suppresses transaction costs to 0.97% of starting capital and achieves the shallowest maximum drawdown (-16.04%). However, learned risk-aversion leads the agents to hold conservative cash buffers, resulting in cash drag against a fully-invested passive Equal-Weight benchmark during an extended bull market.

Keywords: Deep Reinforcement Learning, Portfolio Management, Proximal Policy Optimisation, Dirichlet Policy, Differential Sharpe Ratio, Turnover Regularisation, POMDP, Walk-Forward Validation.

Contents

Acknowledgement and Abstract **iii**

1 Introduction **1**
1.1 Background and Motivation . . . 1
1.2 Research Objectives . . . 1
1.3 Research Questions . . . 1

2 Literature Review **2**

3 Research Gap and Problem Formulation **3**
3.1 Identified Research Gap . . . 3
3.2 POMDP Formulation of Daily Portfolio Allocation . . . 4

4 Dataset and Experimental Methodology **5**
4.1 Equity Universe and Data Acquisition . . . 5
4.2 Feature Engineering and Market Indicators . . . 5
4.3 State Representation and Temporal Sequences . . . 7
4.4 Walk-Forward Evaluation Protocol . . . 7

5 Model Architecture and Mathematical Formulation **7**
5.1 Overview of the Actor–Critic Architecture . . . 7
5.2 State Encoders: MLP versus Recurrent LSTM . . . 8
5.3 Continuous Dirichlet Policy on the Simplex . . . 9
5.4 Portfolio Dynamics and Transaction Cost Accounting . . . 10
5.5 Model Reward Formulations (M1 to M4) . . . 10
5.6 PPO Policy Optimisation . . . 11

6 Controlled Ablation Study Design **11**
6.1 Methodological Rationale for Controlled Ablation . . . 11
6.2 Specification of Ablation Models M1–M4 . . . 11
6.3 Control Invariants and Verification . . . 13

7 Results and Discussion **13**
7.1 Overall Out-of-Sample Performance . . . 13
7.2 Analysis of Ablation Steps . . . 14
7.3 Fold-by-Fold Performance Dynamics . . . 15
7.4 Equal-Weight Benchmark Outperformance and Gap Decomposition . . . 16
7.5 M4 Hyperparameter Tuning and Validation Dynamics . . . 17

8 Limitations and Future Work **17**
8.1 Methodological and Empirical Limitations . . . 17
8.2 Directions for Future Research . . . 18

9 Conclusion **18**
9.1 Summary of Research Findings . . . 18

9.2 Answers to Research Questions 19
9.3 Concluding Remarks 19

Chapter 1 — Introduction

1.1 Background and Motivation

Financial markets represent notoriously difficult environments for autonomous sequential decision-making. Daily asset price movements are characterised by low signal-to-noise ratios, non-stationary underlying return distributions, and partial observability of the true macroeconomic and order-flow state. An algorithmic trading agent observing historical prices and volumes never directly observes the complete market environment, but only a noisy, lossy projection of it.

Portfolio allocation is an intrinsically sequential problem: the economic outcome of an allocation decision depends critically on the history of prior allocations and on how portfolio positions are actively rebalanced over time, rather than on isolated, one-step directional forecasts. This sequential nature makes Deep Reinforcement Learning (DRL) a natural paradigm for portfolio management, as DRL agents learn to optimize cumulative, long-horizon risk-adjusted rewards under market uncertainty rather than one-step predictive accuracy.

However, practical algorithmic execution requires rigorous accounting for transaction frictions. DRL agents operating in continuous action spaces frequently exhibit excessive reallocation or churning: policies that appear superficially profitable before transaction fees can incur severe losses once realistic transaction costs are subtracted from portfolio wealth. While extensive prior literature has separately proposed temporal neural architectures [7], risk-aware reward objectives [3, 4], and dynamic walk-forward evaluation protocols [2], these mechanisms are almost never isolated within a single, consistent experimental environment. Differences in datasets, transaction cost models, feature sets, and evaluation frameworks across published studies make it nearly impossible to discern whether reported empirical gains arise from the neural architecture, reward formulation, turnover suppression, or benign testing conditions.

1.2 Research Objectives

To address this limitation, the specific objectives of this B.Tech. project are as follows:

1. **Controlled Four-Stage Ablation:** To design and implement a single-factor controlled four-stage ablation framework—Model M1: PPO (MLP), Model M2: LSTM-PPO, Model M3: LSTM-PPO with Differential Sharpe Ratio (DSR) reward, and Model M4: LSTM-PPO with DSR and turnover regularisation—that isolates the incremental impact of temporal memory, risk-aware reward shaping, and turnover control on a continuous, Dirichlet-parameterised portfolio allocation policy under identical data, cost, and training conditions.

2. **Unified POMDP Environment and Protocol:** To define a single, rigorous Partially Observable Markov Decision Process (POMDP) trading environment incorporating causal feature engineering, exact proportional transaction cost accounting, and a strict chronological walk-forward evaluation protocol across eight years of Indian equity data (2018–2025).

3. **Multi-Dimensional Empirical Evaluation:** To evaluate all models across a comprehensive set of cost-aware, risk-adjusted metrics—including cumulative profit, annualized return, Sharpe ratio, Sortino ratio, maximum drawdown, turnover, and cumulative transaction costs—to determine which mechanisms genuinely contribute to robust out-of-sample performance.

1.3 Research Questions

This thesis investigates and answers four primary research questions:

- **RQ1 (Temporal Architecture):** Does recurrent temporal modeling via an LSTM encoder improve out-of-sample risk-adjusted returns and policy stability compared to a static MLP baseline?
- **RQ2 (Risk-Aware Reward Shaping):** Does replacing raw net portfolio returns with the recursive online Differential Sharpe Ratio (DSR) improve Sharpe ratios and reduce maximum drawdowns?
- **RQ3 (Turnover Regularisation):** Does augmenting the reward with an explicit quadratic turnover penalty effectively suppress excessive trading churn and transaction costs without destroying wealth accumulation?
- **RQ4 (Benchmark Competitiveness):** Can a continuous actor-critic DRL agent outperform a passive Equal-Weight benchmark in a realistic, friction-constrained Indian equity market?

Chapter 2 — Literature Review

The application of deep reinforcement learning (DRL) to financial trading and portfolio management has expanded rapidly, yet critical methodological challenges persist regarding comparability, state representations, reward formulations, and action parameterisations.

Millea [3] highlights extensive methodological inconsistency across quantitative DRL trading research, noting that disparate studies evaluate policies using heterogeneous asset universes, arbitrary pre-processing pipelines, divergent friction assumptions, and non-standardised evaluation protocols, rendering direct cross-study comparisons highly unreliable. The survey emphasizes that many published models claim algorithmic superiority primarily due to favorable dataset periods or unmodelled execution costs. Furthermore, Millea [3] discusses risk-sensitive reward formulations such as the Differential Sharpe Ratio (DSR), which computes an online, recursive estimate of risk-adjusted return by tracking first and second return moments, penalising variance rather than merely rewarding raw nominal returns. These insights underscore the necessity of a controlled, single-environment evaluation framework.

In the domain of neural state representation and temporal feature extraction, Zou et al. [7] implement a cascaded LSTM-PPO architecture for automated stock trading. Their empirical results demonstrate that historical sequential observations are crucial for capturing multi-period market momentum and temporal dependencies that static networks fail to perceive. Through exhaustive time-window and hidden-size experiments, Zou et al. [7] show that a lookback horizon of $W = 30$ trading days paired with an LSTM hidden dimension of $H = 512$ provides an effective temporal summary of price dynamics. However, their policy reward remains strictly return-based, leaving open the question of how risk-adjusted reward shaping interacts with recurrent feature representations under identical trading frictions.

Investigating reward engineering, Huang et al. [1] propose a self-rewarding reinforcement learning mechanism that combines expert-crafted trading rules with a learned internal reward network for strategy optimization. While their findings demonstrate that reward shaping substantially influences learned policy behavior, their approach relies on complex auxiliary networks rather than mathematically derived, risk-adjusted reward signals like the Differential Sharpe Ratio evaluated under identical architectural conditions.

Regarding empirical evaluation rigor, Liu et al. [2] introduce dynamic market datasets and a rolling train-validation-test framework designed to benchmark financial reinforcement learning algorithms. Their framework addresses survivorship bias and non-stationary market regimes by continuously updating training and evaluation windows chronologically. While Liu et al. [2] significantly elevates evaluation methodology in algorithmic trading, their benchmarks do not isolate the individual effects

of temporal memory, risk-sensitive rewards, and turnover regularisation via single-factor controlled ablations.

Addressing market regime sensitivity, Wang and Liu [4] propose an adaptive risk-sensitive reinforcement learning framework for portfolio optimization across shifting market conditions, reporting notable improvements in downside risk mitigation. Concurrently, their architecture incorporates risk constraints and continuous allocation; however, their experimental setup modifies the state encoder, reward formulation, and execution constraints simultaneously, preventing the precise attribution of empirical gains to any single design component.

Finally, action space formulation represents a fundamental mathematical consideration in portfolio management. The portfolio weight vector must strictly lie on the unit simplex: $\sum_{i=1}^{N+1} w_i = 1$ with $w_i \geq 0$. While many existing RL studies apply an ad-hoc Softmax function over unconstrained Gaussian outputs, Yang et al. [6] demonstrate that parameterising the continuous actor directly via a multivariate Dirichlet distribution mathematically guarantees simplex constraints without heuristic clipping or distorted probability densities. By employing Dirichlet policy distributions within an off-policy reinforcement learning framework, Yang et al. [6] establish that exact analytical likelihood evaluation over the simplex manifold enhances gradient stability and portfolio efficiency.

Chapter 3 — Research Gap and Problem Formulation

3.1 Identified Research Gap

A critical synthesis of the literature reveals that while individual advancements have been made in temporal neural architectures, risk-aware reward design, and realistic evaluation methods, these mechanisms have consistently been studied in isolation across disparate datasets, conflicting trading friction assumptions, and ad-hoc architectures.

Main Research Gap: Among the studies reviewed, we did not identify a controlled empirical framework that systematically isolates *temporal memory*, *risk-aware reward shaping*, and *turnover regularisation* under identical market data, feature representations, transaction-cost models, training budgets, and chronological out-of-sample walk-forward evaluations. Without isolating these mechanisms within a single-factor ablation framework, it is impossible to determine whether reported performance improvements stem from the sequence encoder, the reward signal, turnover suppression, or simply idiosyncratic differences in experimental setups.

Table 3.1 positions this thesis relative to the reviewed benchmark literature, highlighting the comprehensive and controlled nature of our design.

Table 3.1: Positioning of this work relative to the reviewed literature.

Paper	Temporal Memory	Risk-Aware / Reward	Turnover Control	Continuous Allocation	Controlled Ablation
Millea (2021) [3]		✓ *(risk-aware rewards discussed)*			
Zou et al. (2023) [7]	✓				
Huang et al. (2024) [1]		✓			
Liu et al. (2024) [2]				✓ *(portfolio weights)*	
Wang & Liu (2025) [4]		✓ *(risk-sensitive reward)*	✓	✓	
Yang et al. (2022) [6]				✓ *(Dirichlet distribution)*	
This work	✓	✓ *(DSR)*	✓ *(turnover penalty)*	✓	✓

3.2 POMDP Formulation of Daily Portfolio Allocation

We formulate daily portfolio allocation as a Partially Observable Markov Decision Process (POMDP), defined by the 7-tuple:

$$\mathcal{P} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle, \tag{3.1}$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{T}$ represents transition dynamics, $\mathcal{R}$ is the reward function, Ω is the observation space, $\mathcal{O}$ is the observation probability distribution, and $\gamma \in [0, 1)$ is the discount factor.

3.2.1 Why Daily Financial Trading is a POMDP

In standard reinforcement learning, a Markov Decision Process assumes the Markov property: the future state depends solely on the current state and action, $\mathbb{P}(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) = \mathbb{P}(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \ldots, \mathbf{s}_0)$. In financial markets, this assumption fails when applied directly to observable market indicators:

1. **Unobservable True State $\mathcal{S}$:** The complete economic state of the financial system encompasses global macroeconomic variables, institutional limit order book depth, private liquidity flows, insider order placement, and market participant psychology. These state variables are inherently unobservable to an algorithmic agent operating from public exchange feeds.

2. **Partial and Noisy Observation Ω:** The agent perceives only an observation vector $\mathbf{o}_t \in \Omega$ generated via the observation model $\mathcal{O}(\mathbf{o}_t \mid \mathbf{s}_t)$, consisting of daily open, high, low, close prices, trading volumes, and derived technical indicators.

3. **Non-Markovian Indicator Dynamics:** An identical instantaneous technical indicator vector (e.g., RSI = 65, MACD crossover) can lead to vastly different subsequent price trajectories depending on latent macroeconomic conditions (e.g., an early bull breakout vs. an institutional distribution top).

Because instantaneous observations $\mathbf{o}_t$ do not satisfy the Markov property, the agent cannot make optimal decisions based solely on a single-day snapshot. Instead, it must construct an internal belief state $b(\mathbf{s}_t)$ over the history of observations $\mathbf{F}_t = [\mathbf{o}_{t-W+1}, \ldots, \mathbf{o}_t]$. In this work, recurrent neural networks (LSTM) are employed to learn this internal temporal summary $\mathbf{h}_t \approx b(\mathbf{s}_t)$ end-to-end, directly mitigating partial observability.

3.2.2 Action Space $\mathcal{A}$ and Simplex Manifold

The portfolio consists of $N = 10$ risky equities and 1 risk-free cash account, yielding an action dimension of $M = N + 1 = 11$. The action vector $\mathbf{a}_t = [a_{t,1}, \ldots, a_{t,N+1}]^\top$ represents target portfolio weights, strictly constrained to the standard unit simplex:

$$\mathcal{A} \equiv \Delta^N = \left\{ \mathbf{w}_t \in \mathbb{R}^{N+1} \;\middle|\; w_{i,t} \geq 0 \; \forall i, \quad \sum_{i=1}^{N} w_{i,t} + w_{\text{cash},t} = 1 \right\}. \tag{3.2}$$

Following Yang et al. [6], this constraint is enforced directly by parameterising the actor network over a multivariate Dirichlet distribution, eliminating the need for heuristic projection or unconstrained Gaussian clipping.

Chapter 4 — Dataset and Experimental Methodology

4.1 Equity Universe and Data Acquisition

The experimental stock universe comprises $N = 10$ prominent large-cap equities actively traded on the National Stock Exchange of India (NSE), spanning key economic sectors:

1. **RELIANCE.NS** (Energy and Petrochemicals / Conglomerate)
2. **TCS.NS** (Information Technology Services)
3. **HDFCBANK.NS** (Private Commercial Banking)
4. **ICICIBANK.NS** (Private Commercial Banking)
5. **ITC.NS** (Consumer Goods and Diversified Conglomerate)
6. **HINDUNILVR.NS** (Fast-Moving Consumer Goods)
7. **LT.NS** (Engineering and Infrastructure Construction)
8. **SUNPHARMA.NS** (Pharmaceuticals and Healthcare)
9. **MARUTI.NS** (Automotive Manufacturing)
10. **BHARTIARTL.NS** (Telecommunications and Digital Services)

In addition to these ten equities, the portfolio includes an explicit risk-free cash asset (Asset 11), providing the agent with the flexibility to hold liquidity and de-risk during turbulent market regimes.

Daily split- and dividend-adjusted Open, High, Low, Close, and Volume (OHLCV) series were acquired from Yahoo Finance spanning 1 January 2009 to 31 December 2025. Data from 2009–2010 serves as an initial burn-in period to initialize 200-day exponential moving averages, ensuring that all technical indicators are fully settled without boundary artifacts by 1 January 2011. The active dataset comprises 3,699 synchronized trading days.

4.2 Feature Engineering and Market Indicators

For each asset $i \in \{1, \ldots, N\}$ on trading day t, a compact feature vector $\mathbf{f}_{i,t} \in \mathbb{R}^{13}$ is extracted using strictly causal market and technical price transformations. To ensure cross-asset stationarity, all features are normalized by prevailing prices or rolling averages. The mathematical definitions of the thirteen indicators are as follows:

1. **Daily Log Return ($f_{1,i,t}$):** Captures continuous compounding price changes:

$$r_{i,t}^{\log} = \ln\left(\frac{C_{i,t}}{C_{i,t-1}}\right). \tag{4.1}$$

2. **Close-to-Open Relative Return ($f_{2,i,t}$):** Reflects intraday price momentum relative to market open:

$$o_{i,t}^{\text{rel}} = \frac{C_{i,t}}{O_{i,t}} - 1. \tag{4.2}$$

3. **High-to-Low Relative Spread ($f_{3,i,t}$):** Measures intraday trading range and local volatility:

$$hl_{i,t}^{\text{rel}} = \frac{H_{i,t}}{L_{i,t}} - 1. \tag{4.3}$$

4. **Normalized Relative Volume ($f_{4,i,t}$):** Measures trading volume deviation relative to its 20-day Simple Moving Average (SMA):

$$v_{i,t}^{\text{rel}} = \ln\left(\frac{V_{i,t}}{\text{SMA}_{20}(V_{i,t})}\right), \quad \text{where } \text{SMA}_n(V_t) = \frac{1}{n}\sum_{k=0}^{n-1} V_{t-k}. \tag{4.4}$$

5. **50-day EMA Relative Deviation ($f_{5,i,t}$):** Gauges medium-term trend direction. The Exponential Moving Average (EMA) with smoothing factor $\alpha_n = \frac{2}{n+1}$ updates as $\text{EMA}_n(C)_t = \alpha_n C_t + (1-\alpha_n)\text{EMA}_n(C)_{t-1}$:

$$ema_{50,i,t}^{\text{rel}} = \frac{C_{i,t}}{\text{EMA}_{50}(C_{i,t})} - 1. \tag{4.5}$$

6. **200-day EMA Relative Deviation ($f_{6,i,t}$):** Gauges long-term macro trend alignment:

$$ema_{200,i,t}^{\text{rel}} = \frac{C_{i,t}}{\text{EMA}_{200}(C_{i,t})} - 1. \tag{4.6}$$

7. **Relative Strength Index ($f_{7,i,t}$):** Wilder's 14-period RSI measures price velocity and momentum. Let $U_t = \max(C_t - C_{t-1}, 0)$ and $D_t = \max(C_{t-1} - C_t, 0)$. With smoothed averages $\bar{U}_t = \alpha_{14}U_t + (1-\alpha_{14})\bar{U}_{t-1}$ and $\bar{D}_t = \alpha_{14}D_t + (1-\alpha_{14})\bar{D}_{t-1}$:

$$\text{RS}_{i,t} = \frac{\bar{U}_{i,t}}{\bar{D}_{i,t} + 10^{-8}}, \quad \text{RSI}_{i,t} = 100 - \frac{100}{1+\text{RS}_{i,t}}, \quad f_{7,i,t} = \frac{\text{RSI}_{i,t}}{100} - 0.5. \tag{4.7}$$

8. **Moving Average Convergence Divergence ($f_{8,i,t}$):** Normalized trend divergence:

$$\text{MACD}_{i,t} = \frac{\text{EMA}_{12}(C_{i,t}) - \text{EMA}_{26}(C_{i,t})}{C_{i,t}}. \tag{4.8}$$

9. **MACD Signal Line ($f_{9,i,t}$):** Normalized moving average of MACD:

$$\text{Signal}_{i,t} = \frac{\text{EMA}_9\left(\text{EMA}_{12}(C_{i,t}) - \text{EMA}_{26}(C_{i,t})\right)}{C_{i,t}}. \tag{4.9}$$

10. **Average True Range ($f_{10,i,t}$):** Normalized price volatility. The True Range is $\text{TR}_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$:

$$\text{ATR}_{i,t}^{\text{rel}} = \frac{\text{SMA}_{14}(\text{TR}_{i,t})}{C_{i,t}}. \tag{4.10}$$

11. **Bollinger Band %B ($f_{11,i,t}$):** Quantifies price location relative to 2-standard-deviation volatility envelopes around the 20-day SMA ($\text{MB}_t = \text{SMA}_{20}(C_t)$), where $\text{UB}_t = \text{MB}_t + 2\sigma_{20,t}$ and $\text{LB}_t = \text{MB}_t - 2\sigma_{20,t}$:

$$\%B_{i,t} = \frac{C_{i,t} - \text{LB}_{i,t}}{\text{UB}_{i,t} - \text{LB}_{i,t} + 10^{-8}}. \tag{4.11}$$

12. **Bollinger Bandwidth ($f_{12,i,t}$):** Measures cyclical volatility compression and expansion:

$$\text{BW}_{i,t} = \frac{\text{UB}_{i,t} - \text{LB}_{i,t}}{\text{MB}_{i,t}}. \tag{4.12}$$

13. **Average Directional Index ($f_{13,i,t}$):** Wilder's 14-period ADX measures non-directional trend strength:

$$\text{ADX}_{i,t}^{\text{norm}} = \frac{\text{ADX}_{14,i,t}}{100} \in [0,1]. \tag{4.13}$$

4.3 State Representation and Temporal Sequences

The instantaneous market observation vector combines the 13 features across all $N = 10$ stocks, concatenated with the agent's current portfolio weight vector $\mathbf{w}_t^{\text{cur}} \in \mathbb{R}^{N+1}$ (including cash):

$$\mathbf{x}_t = \text{concat}\left(\mathbf{f}_{1,t}, \ldots, \mathbf{f}_{N,t}\right) \in \mathbb{R}^{13N} = \mathbb{R}^{130}, \tag{4.14}$$

$$\mathbf{s}_t = \left[\mathbf{x}_t^\top,\ (\mathbf{w}_t^{\text{cur}})^\top\right]^\top \in \mathbb{R}^{14N+1} = \mathbb{R}^{141}. \tag{4.15}$$

To furnish the agent with temporal context, a sliding historical lookback window of length $W = 30$ trading days is maintained:

$$\mathbf{F}_t = \left[\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t\right]^\top \in \mathbb{R}^{W \times 141}. \tag{4.16}$$

All features use strictly causal backward-looking operators. Feature standardization is performed strictly using the training set statistics $\boldsymbol{\mu}_{\text{train}}$ and $\boldsymbol{\sigma}_{\text{train}}$ of each fold, with values clipped to $[-5.0, 5.0]$ to guard against outliers.

4.4 Walk-Forward Evaluation Protocol

Following the financial machine learning evaluation framework of Liu et al. [2], models are trained and tested using an 8-year chronological rolling-origin walk-forward protocol. Four distinct folds are constructed with 6-year training, 1-year validation, and 2-year held-out test windows, with the origin rolling forward by 2 years per fold (Table 4.1).

Table 4.1: Chronological specification of the four walk-forward folds.

Fold	Training Window (6y)	Validation Window (1y)	Test Window (2y)	Train Steps	Test Steps
1	2011-01-01 to 2016-12-31	2017-01-01 to 2017-12-31	2018-01-01 to 2019-12-31	1,478	489
2	2013-01-01 to 2018-12-31	2019-01-01 to 2019-12-31	2020-01-01 to 2021-12-31	1,475	496
3	2015-01-01 to 2020-12-31	2021-01-01 to 2021-12-31	2022-01-01 to 2023-12-31	1,476	495
4	2017-01-01 to 2022-12-31	2023-01-01 to 2023-12-31	2024-01-01 to 2025-12-31	1,479	494

The test windows are contiguous and non-overlapping, spanning 2018 to 2025 without look-ahead contamination. To ensure statistical reliability, each experiment is trained across three fixed random seeds ($S \in \{0, 1, 2\}$). Policy checkpoints are selected based on the highest validation Sharpe ratio and evaluated once on the out-of-sample test window.

Chapter 5 — Model Architecture and Mathematical Formulation

5.1 Overview of the Actor–Critic Architecture

The portfolio management policy is trained using Proximal Policy Optimisation (PPO) [5] formulated within an Actor–Critic framework. The system architecture decouples temporal representation learning, continuous allocation policy parameterisation on the simplex manifold, and state-value baseline estimation:

1. An **Encoder** processes the state observation to generate a compact 512-dimensional latent representation $\mathbf{h}_t \in \mathbb{R}^{512}$.

2. An **Actor head** maps $\mathbf{h}_t$ to the parameters of a multivariate Dirichlet distribution that outputs portfolio allocations across $M = 11$ assets (10 equities plus cash).

3. A **Critic head** maps $\mathbf{h}_t$ to a scalar baseline estimate $V_\phi(\mathbf{s}_t) \in \mathbb{R}$ approximating the expected cumulative discounted future reward.

Figure 5.1 illustrates the end-to-end information flow, and Figure 5.2 portrays the rollout interaction and PPO optimization pipeline.

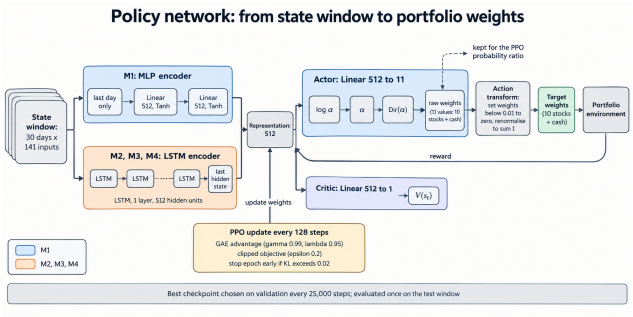


Figure 5.1: Policy architecture from state to portfolio weights.

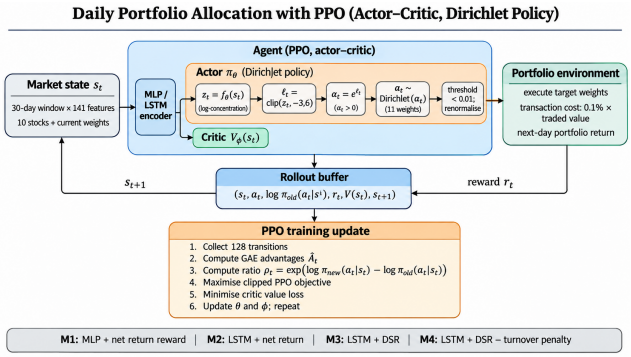


Figure 5.2: PPO Dirichlet allocation and environment loop.

5.2 State Encoders: MLP versus Recurrent LSTM

To rigorously evaluate the utility of temporal memory, two alternative encoder backbones are examined:

5.2.1 Multi-Layer Perceptron (MLP) Encoder (Model M1)

The feedforward MLP baseline observes solely the instantaneous observation vector at day t, $\mathbf{x}_t = \mathbf{s}_t \in \mathbb{R}^{141}$, discarding multi-period historical trajectories. The representation $\mathbf{h}_t$ is computed via two fully-connected layers with Tanh activations:

$$\mathbf{h}_t = \tanh\left(\mathbf{W}_2 \tanh\left(\mathbf{W}_1 \mathbf{x}_t + \mathbf{b}_1\right) + \mathbf{b}_2\right), \tag{5.1}$$

where $\mathbf{W}_1 \in \mathbb{R}^{512\times 141}$ and $\mathbf{W}_2 \in \mathbb{R}^{512\times 512}$.

5.2.2 Recurrent LSTM Encoder (Models M2–M4)

To capture sequential price momentum and multi-period dependencies, Models M2, M3, and M4 pass the sliding history window $\mathbf{F}_t = [\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t] \in \mathbb{R}^{W\times 141}$ ($W = 30$) through a recurrent Long Short-Term Memory (LSTM) network:

$$(\mathbf{h}_t, \mathbf{c}_t) = \text{LSTM}(\mathbf{F}_t), \tag{5.2}$$

with hidden representation size $H = 512$, following the empirical findings of Zou et al. [7]. The LSTM is trained end-to-end with the policy and value networks; rather than serving as an explicit price forecaster, $\mathbf{h}_t$ serves as a learned temporal belief state for sequential allocation.

5.3 Continuous Dirichlet Policy on the Simplex

A fundamental challenge in continuous portfolio control is ensuring that allocated asset weights satisfy the portfolio simplex constraints:

$$w_{i,t}^{\text{target}} \geq 0, \quad w_{\text{cash},t}^{\text{target}} \geq 0, \quad \sum_{i=1}^{N} w_{i,t}^{\text{target}} + w_{\text{cash},t}^{\text{target}} = 1. \tag{5.3}$$

Following Yang et al. [6], the policy is parameterised using a multivariate Dirichlet distribution, which natively defines a probability measure directly over the open unit simplex without requiring arbitrary heuristic projections.

5.3.1 Actor Mapping to Dirichlet Parameters

The actor network maps the latent representation $\mathbf{h}_t$ to an unconstrained linear vector $\mathbf{z}_t \in \mathbb{R}^{N+1}$:

$$\mathbf{z}_t = \mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a, \quad \mathbf{W}_a \in \mathbb{R}^{(N+1)\times 512}, \ \mathbf{b}_a \in \mathbb{R}^{N+1}. \tag{5.4}$$

To ensure numerical stability during gradient backpropagation, the outputs are clamped to a safe log-concentration range and exponentiated to yield strictly positive Dirichlet concentration parameters $\boldsymbol{\alpha}_t \in \mathbb{R}_{>0}^{N+1}$:

$$\boldsymbol{\ell}_t = \text{clip}\left(\mathbf{z}_t, -3.0, 6.0\right), \tag{5.5}$$

$$\alpha_{i,t} = \exp\left(\ell_{i,t}\right) \in \left[e^{-3}, e^{6}\right] \approx [0.0498, 403.43]. \tag{5.6}$$

Clamping prevents underflow (which leads to degenerate distributions concentrated at vertices) and overflow (which causes numerical instability in the Gamma function derivative).

5.3.2 Action Sampling and Expected Allocation

The continuous action $\mathbf{a}_t \in \Delta^N$ is sampled from the Dirichlet probability density:

$$\pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t) = \frac{\Gamma\left(\sum_{i=1}^{N+1} \alpha_{i,t}\right)}{\prod_{i=1}^{N+1} \Gamma(\alpha_{i,t})} \prod_{i=1}^{N+1} a_{i,t}^{\alpha_{i,t}-1}, \tag{5.7}$$

where $\Gamma(z)$ is the standard Gamma function. Sampling is performed via normalized independent Gamma variates: $g_{i,t} \sim \text{Gamma}(\alpha_{i,t}, 1)$ with $a_{i,t} = g_{i,t} / \sum_j g_{j,t}$. The expected portfolio allocation is:

$$\mathbb{E}\left[a_{i,t}\right] = \frac{\alpha_{i,t}}{\sum_{j=1}^{N+1} \alpha_{j,t}}. \tag{5.8}$$

During deterministic out-of-sample evaluation, the policy executes this expected allocation directly.

5.3.3 Sparsity Thresholding and Renormalisation

Because a Dirichlet distribution with positive parameters assigns non-zero mass across all assets, raw samples can contain minuscule allocations (e.g., 0.05%) that induce unneeded rebalancing fees. A threshold $\tau = 0.01$ (1%) is applied identically across all models:

$$\tilde{w}_{i,t} = \begin{cases} a_{i,t}, & \text{if } a_{i,t} \geq \tau, \\ 0, & \text{if } a_{i,t} < \tau, \end{cases} \qquad w_{i,t}^{\text{target}} = \frac{\tilde{w}_{i,t}}{\sum_{j=1}^{N+1} \tilde{w}_{j,t}}. \tag{5.9}$$

This operation liquidates tiny positions and renormalises the surviving allocations to sum strictly to one.

5.4 Portfolio Dynamics and Transaction Cost Accounting

Let $\mathbf{w}_t^{\text{cur}} \in \Delta^N$ denote the portfolio weights immediately preceding rebalancing at day t. The required reallocation across the risky equities is $\Delta w_{i,t} = w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}}$.

Reallocating between cash and risky assets incurs a proportional transaction fee $c_{\text{trans}} = 0.001$ (10 bps):

$$C_t = c_{\text{trans}} V_{t-1} \sum_{i=1}^{N} |\Delta w_{i,t}| \,. \tag{5.10}$$

End-of-day portfolio wealth V_t evolves according to:

$$V_t = (V_{t-1} - C_t) \left[\sum_{i=1}^{N} w_{i,t}^{\text{target}} (1 + R_{i,t}) + w_{\text{cash},t}^{\text{target}} \right], \tag{5.11}$$

where $R_{i,t} = \frac{C_{i,t}}{C_{i,t-1}} - 1$ is the simple price return. The net portfolio return used throughout the framework is:

$$R_{\text{net},t} = \frac{V_t - V_{t-1}}{V_{t-1}}. \tag{5.12}$$

5.5 Model Reward Formulations (M1 to M4)

The reward signal r_t emitted to the agent constitutes the core ablation dimension:

1. **Models M1 and M2 (Net Return Reward):**

$$r_t = R_{\text{net},t}. \tag{5.13}$$

2. **Model M3 (Differential Sharpe Ratio):** Following Millea [3], an online recursive Differential Sharpe Ratio (DSR) is computed without requiring batch return histories. Moving estimates of the first and second return moments ($A_t \approx \mathbb{E}[R]$ and $B_t \approx \mathbb{E}[R^2]$) update exponentially with adaptation rate $\eta = 0.01$:

$$A_t = A_{t-1} + \eta(R_{\text{net},t} - A_{t-1}), \tag{5.14}$$
$$B_t = B_{t-1} + \eta(R_{\text{net},t}^2 - B_{t-1}). \tag{5.15}$$

   With moment changes $\Delta A_t = A_t - A_{t-1}$ and $\Delta B_t = B_t - B_{t-1}$, the DSR reward is:

$$D_t = \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta B_t}{\left(B_{t-1} - A_{t-1}^2 + \varepsilon_{\text{DSR}}\right)^{3/2}}, \tag{5.16}$$

   where $\varepsilon_{\text{DSR}} = 10^{-8}$ prevents numerical division by zero. For Model M3, $r_t = D_t$.

3. **Model M4 (DSR with Squared Turnover Penalty):** To discourage excessive daily weight churn, an explicit quadratic turnover penalty is subtracted from the DSR reward:

$$r_t = D_t - \lambda_{\text{turn}} \sum_{i=1}^{N} \left(w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}} \right)^2, \tag{5.17}$$

   with $\lambda_{\text{turn}} = 0.1$ selected via validation tuning.

5.6 PPO Policy Optimisation

During environment interaction, the agent collects transitions and stores the raw sampled continuous action $\mathbf{a}_t$ along with its log-likelihood under old parameters θ_{old}:

$$\log \pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{h}_t) = \ln \Gamma\left(\sum_{i=1}^{N+1} \alpha_{i,t}\right) - \sum_{i=1}^{N+1} \ln \Gamma(\alpha_{i,t}) + \sum_{i=1}^{N+1} (\alpha_{i,t} - 1) \ln a_{i,t}. \tag{5.18}$$

Generalized Advantage Estimation (GAE) computes advantages $\hat{A}_t$ using temporal-difference residuals $\delta_t = \tilde{r}_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t)$ with discount $\gamma = 0.99$ and trace decay $\lambda_{\text{GAE}} = 0.95$:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}. \tag{5.19}$$

During mini-batch optimisation, the probability ratio is evaluated for the identical raw action $\mathbf{a}_t$:

$$\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{h}_t)} = \exp\left(\log \pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t) - \log \pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{h}_t)\right). \tag{5.20}$$

The policy parameters θ and critic parameters ϕ are updated by minimising the joint multi-task loss [5]:

$$\mathcal{L}(\theta, \phi) = -\hat{\mathbb{E}}_t \left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] + c_1 \hat{\mathbb{E}}_t \left[(V_\phi(\mathbf{s}_t) - \hat{V}_t^{\text{target}})^2\right] - c_2 \hat{\mathbb{E}}_t[\mathcal{H}(\pi_\theta)], \tag{5.21}$$

with clipping range $\epsilon = 0.2$, critic weight $c_1 = 0.5$, and entropy bonus $c_2 = 0.01$.

Chapter 6 — Controlled Ablation Study Design

6.1 Methodological Rationale for Controlled Ablation

In deep reinforcement learning research, empirical comparisons often suffer from reproducibility crises because novel models introduce multiple architectural alterations, revised reward functions, and distinct hyperparameter tunings simultaneously. Under such confounded experimental designs, it is impossible to determine whether empirical gains arise from neural representational power, risk-adjusted reward shaping, or hyperparameter coincidences.

To establish rigorous scientific causality, this research adopts a strict *single-factor controlled ablation study*. Four model configurations (M1 through M4) are constructed such that each consecutive transition alters exactly one algorithmic mechanism while holding all other parameters, random seeds, training budgets, network widths, transaction costs, and data folds strictly invariant.

6.2 Specification of Ablation Models M1–M4

The four ablation models are defined as follows:

1. **Model M1 (MLP Baseline + Net Return):** Evaluates a two-layer feedforward MLP observing solely the instantaneous market observation $\mathbf{s}_t \in \mathbb{R}^{141}$. The reward is the realised net rate of return ($r_t = R_{\text{net},t}$) without turnover regularisation ($\lambda_{\text{turn}} = 0$). This serves as the static, non-temporal baseline.

2. **Model M2 (LSTM Recurrent Encoder + Net Return):** Replaces the static MLP with a single-layer LSTM network with hidden dimension $H = 512$, processing the sliding 30-day

Algorithm 1: PPO Daily Portfolio Allocation with Dirichlet Policy

Input: Market series $\mathbf{X}$, returns $\mathbf{R}$, cost c_{trans}, threshold τ, penalty λ_{turn}, PPO parameters $\gamma, \lambda_{\text{GAE}}, \epsilon, \eta_{\text{LR}}$.
Initialisation: Actor θ, Critic ϕ, reward normaliser $\mathcal{N}$, initial weights $\mathbf{w}_0^{\text{cur}} \leftarrow [0, \ldots, 0, 1]^\top$ (cash), DSR moments $A \leftarrow 0, B \leftarrow 0$.

for iteration $= 1, 2, \ldots$ until $T_{\text{total}} = 300,000$ steps **do**
  Reset rollout buffer $\mathcal{B} \leftarrow \emptyset$.
  for $t = 1$ to $T_{\text{rollout}} = 128$ **do**
    Compute representation $\mathbf{h}_t \leftarrow \text{Encoder}(\mathbf{s}_t)$ (MLP for M1; LSTM for M2–M4).
    Compute concentrations $\boldsymbol{\ell}_t \leftarrow \text{clip}(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a, -3, 6)$ and $\boldsymbol{\alpha}_t \leftarrow \exp(\boldsymbol{\ell}_t)$.
    Sample raw action variates $g_i \sim \text{Gamma}(\alpha_{i,t}, 1)$, set $\mathbf{a}_t \leftarrow \mathbf{g} / \sum_i g_i$.
    Evaluate old log-likelihood $lp_t \leftarrow \log \pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t)$ and value $V_t \leftarrow V_\phi(\mathbf{h}_t)$.
    Apply threshold $\tau = 0.01$: $\tilde{w}_i \leftarrow a_{i,t} \cdot \mathbf{1}_{a_{i,t} \geq \tau}$, set $\mathbf{w}_t^{\text{target}} \leftarrow \tilde{\mathbf{w}} / \sum_j \tilde{w}_j$.
    Execute trade: compute Turnover_t, cost C_t, net return $R_{\text{net},t}$, new wealth V_t.
    Compute model reward r_t (Net return for M1/M2; DSR for M3; DSR $-\lambda_{\text{turn}} \sum (\Delta w_i)^2$ for M4).
    Store transition $(\mathbf{s}_t, \mathbf{a}_t, lp_t, V_t, r_t)$ in buffer $\mathcal{B}$.
  end for
  Compute GAE advantages $\hat{A}_t$ and return targets $\hat{V}_t^{\text{target}}$.
  for epoch $= 1$ to $K = 10$ **do**
    for each minibatch of size $B = 64$ **do**
      Evaluate current log-likelihoods $\log \pi_\theta(\mathbf{a}_i \mid \mathbf{h}_i)$ and values $V_\phi(\mathbf{h}_i)$.
      Compute probability ratio $\rho_i(\theta) \leftarrow \exp(\log \pi_\theta - lp_i)$ and multi-task loss $\mathcal{L}(\theta, \phi)$.
      Update (θ, ϕ) via Adam optimiser ($\eta_{\text{LR}} = 10^{-4}$) with gradient clipping 0.5.
    end for
  end for
end for

Figure 5.3: Complete training and policy update algorithm for PPO portfolio allocation.

historical window. The reward remains net return ($r_t = R_{\text{net},t}$). Transition M1 $\rightarrow$ M2 strictly isolates the impact of **temporal memory and sequence modeling**.

3. **Model M3 (LSTM Encoder + Differential Sharpe Ratio):** Retains the identical LSTM backbone ($H = 512$), but replaces the net return reward with the recursive online Differential Sharpe Ratio ($r_t = D_t$). Transition M2 $\rightarrow$ M3 strictly isolates the impact of **risk-aware reward engineering** against raw return maximisation.

4. **Model M4 (LSTM + DSR + Squared Turnover Regularisation):** Retains the LSTM backbone and DSR reward, augmented by an explicit quadratic penalty on risky asset rebalancing ($r_t = D_t - \lambda_{\text{turn}} \sum_{j=1}^{N} (w_{j,t}^{\text{target}} - w_{j,t}^{\text{cur}})^2$) with $\lambda_{\text{turn}} = 0.1$. Transition M3 $\rightarrow$ M4 strictly isolates the impact of **explicit turnover regularisation and transaction-friction suppression**.

Table 6.1 summarises the architectural and functional properties of the four configurations.

Table 6.1: Systematic ablation matrix across models M1 to M4.

Configuration Property	**Model M1**	**Model M2**	**Model M3**	**Model M4**
Encoder Backbone	MLP (2-layer)	LSTM (1-layer)	LSTM (1-layer)	LSTM (1-layer)
Temporal Window (W)	1 day (last row)	30 days	30 days	30 days
Latent Dimension (H)	512	512	512	512
Policy Head	Dirichlet Δ^{11}	Dirichlet Δ^{11}	Dirichlet Δ^{11}	Dirichlet Δ^{11}
Action Threshold (τ)	0.01	0.01	0.01	0.01
Reward Formulation	Net Return	Net Return	DSR (D_t)	DSR (D_t)
Turnover Regularisation	None ($\lambda = 0$)	None ($\lambda = 0$)	None ($\lambda = 0$)	L_2 Penalty ($\lambda = 0.1$)
Environment Cost Model	0.1% per turnover	0.1% per turnover	0.1% per turnover	0.1% per turnover
Training Budget	300,000 steps	300,000 steps	300,000 steps	300,000 steps
Ablation Mechanism	*Baseline*	**+ Temporal**	**+ Risk-Aware**	**+ Turnover Penalty**

6.3 Control Invariants and Verification

To guarantee that no hidden variance compromises the ablation comparisons:

1. **Weight Initialisation Parity:** In `src/models.py`, the linear Actor head ($\mathbf{W}_a, \mathbf{b}_a$) and Critic head ($\mathbf{W}_c, \mathbf{b}_c$) are instantiated *before* the encoder network. When initialised under an identical pseudo-random seed, models M1 and M2 possess bitwise identical initial weights in their policy and value heads.

2. **Equal Computational Budget:** All models are trained for exactly 300,000 environment interaction steps with an identical Adam learning rate (10^{-4}), rollout horizon (128), and minibatch size (64).

3. **Identical Market Realisation and Costs:** Each model experiences the exact same price sequences, transaction cost deductions (0.1%), and execution mechanics.

4. **Automated Assertion:** The execution harness enforces `assert_controlled_ablation()` at startup, raising an assertion failure if any model transition modifies more than exactly one algorithmic dimension.

Chapter 7 — Results and Discussion

7.1 Overall Out-of-Sample Performance

Out-of-sample evaluation was conducted across four chronological 2-year test segments spanning 2018 through 2025. Table 7.1 presents the pooled test performance metrics (averaged across four folds and three random seeds) for models M1–M4 alongside the passive Equal-Weight (EW) 1/N benchmark.

Table 7.1: Pooled out-of-sample test performance across 4 folds and 3 seeds (mean ± standard deviation).

Model	Cum. Return (%)	CAGR (%)	Sharpe Ratio	Sortino	Max DD (%)	Cost (% of V_0)
M1 (MLP)	20.62 ± 5.14	9.90 ± 2.31	0.71 ± 0.13	1.03 ± 0.20	-19.50 ± 1.04	9.68 ± 0.10
M2 (LSTM)	28.23 ± 2.03	13.37 ± 0.88	0.92 ± 0.07	1.34 ± 0.09	-18.52 ± 1.35	4.27 ± 1.44
M3 (+ DSR)	26.93 ± 3.27	12.68 ± 1.56	0.92 ± 0.11	1.34 ± 0.17	-19.95 ± 0.81	4.14 ± 1.15
M4 (+ Turn. Pen.)	20.97 ± 4.19	10.09 ± 1.85	0.92 ± 0.17	1.37 ± 0.26	$\mathbf{-16.04 \pm 1.72}$	$\mathbf{0.97 \pm 0.15}$
Equal-Weight	**36.48**	**17.08**	**1.13**	**1.66**	-18.91	**0.61**

Figure 7.1 illustrates cumulative out-of-sample portfolio profit (assuming $V_0 =$ INR 1,000,000 initial capital reinvested sequentially) across the continuous 2018–2025 timeline. Figure 7.2 portrays peak-to-trough drawdowns, and Figure 7.3 depicts profit across each 2-year test window.

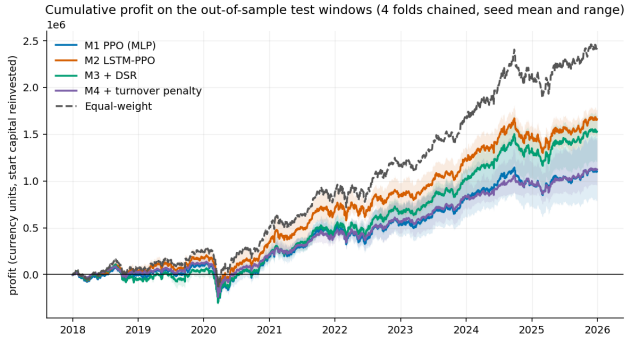


Figure 7.1: Cumulative out-of-sample profit across 4 chained test windows (2018–2025).

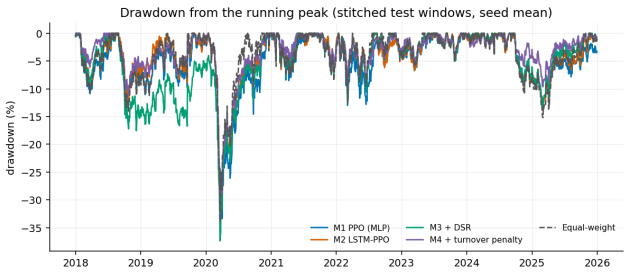


Figure 7.2: Out-of-sample portfolio drawdown from running peak (2018–2025).

7.2 Analysis of Ablation Steps

To establish statistical significance, paired difference tests were computed across all (fold, seed) tuples using 10,000-sample fold-level bootstrap resampling. Table 7.2 reports the resulting paired deltas and 95% confidence intervals.

Table 7.2: Paired ablation differences ($b - a$) with 95% bootstrap confidence intervals.

Ablation Transition	Metric	Mean Delta	95% CI	Folds Improved	Pairs Improved
M1 → M2 (Temporal Memory)	Sharpe Ratio	+0.201	[+0.156, +0.246]	4/4	9/12
	Cum. Return (%)	+7.61%	[+4.81%, +10.01%]	4/4	10/12
	Daily Turnover	−0.106	[−0.138, −0.070]	4/4	11/12
	Cost (% of V_0)	−5.41%	[−7.37%, −3.34%]	4/4	11/12
M2 → M3 (DSR Reward)	Sharpe Ratio	+0.003	[−0.295, +0.222]	3/4	7/12
	Cum. Return (%)	−1.30%	[−9.33%, +4.45%]	2/4	6/12
	Max Drawdown (%)	−1.43%	[−5.36%, +0.81%]	3/4	6/12
	Daily Turnover	−0.002	[−0.019, +0.014]	2/4	7/12
M3 → M4 (Turnover Penalty)	Sharpe Ratio	+0.005	[−0.236, +0.280]	2/4	6/12
	Cum. Return (%)	−5.96%	[−13.29%, +3.37%]	1/4	3/12
	Max Drawdown (%)	**+3.90%**	[+1.45%, +7.00%]	**4/4**	**10/12**
	Daily Turnover	**−0.060**	[−0.082, −0.044]	**4/4**	**12/12**
	Cost (% of V_0)	**−3.17%**	[−4.01%, −2.43%]	**4/4**	**12/12**
M1 → M4 (All Combined)	Sharpe Ratio	+0.209	[+0.012, +0.386]	3/4	8/12
	Max Drawdown (%)	+3.45%	[+2.52%, +4.03%]	4/4	8/12
	Daily Turnover	−0.167	[−0.190, −0.144]	4/4	12/12
	Cost (% of V_0)	−8.71%	[−10.07%, −7.27%]	4/4	12/12

7.2.1 Step 1: Impact of Temporal Memory (M2 versus M1)

Transitioning from the MLP (M1) to the LSTM (M2) constitutes the single most consequential enhancement in the ablation hierarchy. The Sharpe ratio rises significantly from 0.714 to 0.915 (+0.201,

95% CI [+0.156, +0.246]), improving across all four folds (4/4). Simultaneously, daily turnover plummets by more than half, from 0.186 to 0.080, reducing total transaction costs from 9.68% to 4.27% of starting capital.

The MLP observes only the current day's normalized state, causing it to react erratically to day-to-day market noise. In contrast, the internal memory cells of the LSTM smooth out short-term oscillations and track persistent medium-term trends, generating substantial execution savings.

7.2.2 Step 2: Impact of Differential Sharpe Ratio (M3 versus M2)

Replacing raw net return with the Differential Sharpe Ratio (M3) yields an incremental Sharpe delta of +0.003 ([−0.295, +0.222]). While M3 maintains high Sharpe ratios in Folds 3 and 4 (1.328 and 1.164), its out-of-sample return is slightly lower than M2 (26.93% vs. 28.23%). The recursive exponential moving average of DSR dampens aggressive rebalancing during trending bull markets, curtailing profit capture while modestly stabilising risk metrics.

7.2.3 Step 3: Impact of Turnover Regularisation (M4 versus M3)

Incorporating the quadratic turnover penalty in Model M4 produces a decisive shift in trading behaviour:

- **Drastic Cost Reduction:** Average daily turnover drops from 0.079 to 0.019 (a 76% reduction), bringing total transaction costs down to 0.97% of starting capital, which closely approaches the passive benchmark (0.61%).
- **Superior Downside Protection:** Max drawdown improves by +3.90% (from −19.95% to −16.04%, 95% CI [+1.45%, +7.00%]), achieving the shallowest drawdown among all models, including Equal-Weight (−18.91%).
- **Trading Frictions and Turnover:** Figure 7.4 explicitly demonstrates the collapse in trading activity and transaction friction across the ablation sequence.

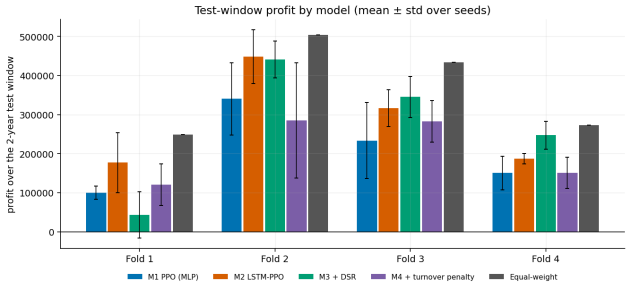


Figure 7.3: Profit per 2-year test window by model.

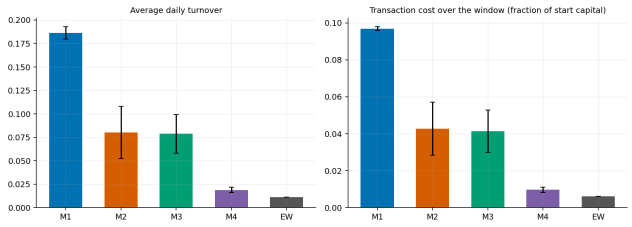


Figure 7.4: Average daily turnover and transaction costs.

7.3 Fold-by-Fold Performance Dynamics

Table 7.3 provides the breakdown of cumulative returns across each fold. Performance exhibits marked regime dependence:

- **Fold 1 (2018–2019):** Moderate growth environment. All RL models generate positive return (10.0% to 17.8%), though trailing EW (24.9%).
- **Fold 2 (2020–2021):** Highly volatile pandemic crash and subsequent massive recovery. M2 and M3 capture dramatic gains (44.8% and 44.1%), whereas M4 lagged (28.5%) because its turnover penalty discouraged rapid capital reallocation during the rapid post-crash rebound.

- **Fold 3 (2022–2023):** Inflationary cycle and sector rotation. M3 achieves its strongest risk-adjusted performance with a Sharpe ratio of 1.328, closely tracking EW (34.5% vs. 43.4%).
- **Fold 4 (2024–2025):** Secular bull market. M3 delivers 24.7% return, outperforming M1 and M4.

Table 7.3: Out-of-sample cumulative return (%) per walk-forward fold (mean ± standard deviation over seeds).

Fold Window	Model M1	Model M2	Model M3	Model M4	Equal-Weight
Fold 1 (2018–2019)	10.0 ± 1.7	17.8 ± 7.7	4.4 ± 5.9	12.1 ± 5.3	**24.9**
Fold 2 (2020–2021)	34.0 ± 9.2	44.8 ± 6.9	44.1 ± 4.7	28.5 ± 14.8	**50.4**
Fold 3 (2022–2023)	23.4 ± 9.7	31.6 ± 4.7	34.5 ± 5.2	28.3 ± 5.3	**43.4**
Fold 4 (2024–2025)	15.0 ± 4.2	18.7 ± 1.3	24.7 ± 3.6	15.1 ± 4.0	**27.3**
Pooled Mean	20.6 ± 5.1	28.2 ± 2.0	26.9 ± 3.3	21.0 ± 4.2	**36.5**

7.4 Equal-Weight Benchmark Outperformance and Gap Decomposition

In the final experimental test, the passive Equal-Weight benchmark achieved the highest cumulative profit (36.48% pooled return, Sharpe 1.131), outperforming all four RL models. Rather than obscuring this finding, we perform a rigorous return gap decomposition to understand why active RL underperformed 1/N.

Table 7.4 decomposes the cumulative return shortfall into two orthogonal components:

1. **Cash Drag:** The return forfeited by holding an uninvested cash allocation during an overall upward-trending market.
2. **Stock Choice and Timing:** The differential return resulting from active equity weight deviations relative to equal-weighting across the ten equities.

Table 7.4: Decomposition of cumulative out-of-sample return gap against Equal-Weight (% points).

Model	EW Return	Cash Drag	EW Same Cash	Stock Selection	Model Return
M1 (MLP)	135.66	−13.67	121.99	−2.11	119.88
M2 (LSTM)	135.66	−9.99	125.67	−2.14	123.53
M3 (+ DSR)	135.66	−15.14	120.52	−3.32	117.20
M4 (+ Turn. Pen.)	135.66	**−42.35**	93.31	−7.84	**85.47**

Table 7.5 reports the average portfolio allocations across the entire evaluation horizon.

Table 7.5: Average asset allocations across models over the out-of-sample evaluation period.

Model	RELI	TCS	HDFCB	ICICIB	ITC	HIND	LT	SUNP	MARU	BHAR	Cash
M1	0.088	0.120	0.085	0.087	0.084	0.077	0.091	0.087	0.087	0.094	**0.101**
M2	0.098	0.144	0.073	0.091	0.081	0.087	0.082	0.090	0.098	0.071	**0.087**
M3	0.089	0.119	0.073	0.077	0.072	0.079	0.073	0.094	0.107	0.086	**0.131**
M4	0.066	0.103	0.079	0.061	0.049	0.066	0.057	0.071	0.092	0.032	**0.324**
EW	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	**0.000**

The empirical cause of benchmark outperformance is unequivocal:

- Between 2018 and 2025, the Indian equity market experienced a sustained structural bull market. The passive 1/N benchmark remained 100% invested across all ten large-cap stocks at all times, incurring minimal rebalancing friction (0.61%).

- Model M4 learned to allocate an average of **32.4%** of its portfolio to cash to minimize volatility and avoid turnover penalties. This cash buffer caused a massive −42.35% cash drag relative to Equal-Weight. In contrast, stock timing accounted for only −7.84% deficit.
- Thus, M4's lower return was not caused by poor stock picking, but by its risk-averse preference for capital preservation in cash during a high-momentum bull market.

7.5 M4 Hyperparameter Tuning and Validation Dynamics

Figure 7.5 depicts the hyperparameter sweep over $\lambda_{\text{turn}} \in \{0.01, 0.03, 0.1, 0.3\}$ on Fold 1 validation across two random seeds (seeds 0 and 1).

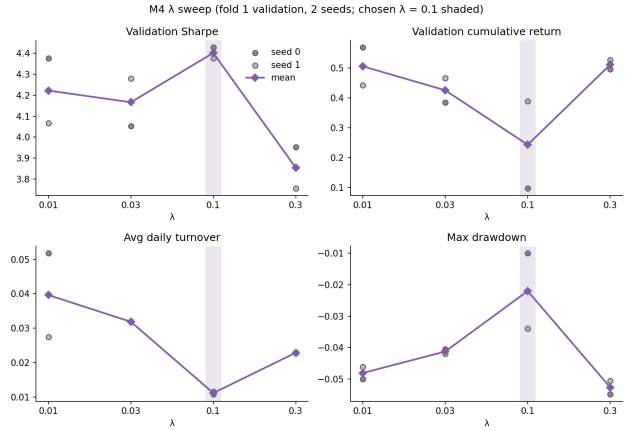


Figure 7.5: M4 turnover penalty parameter sweep (chosen $\lambda = 0.1$ shaded).

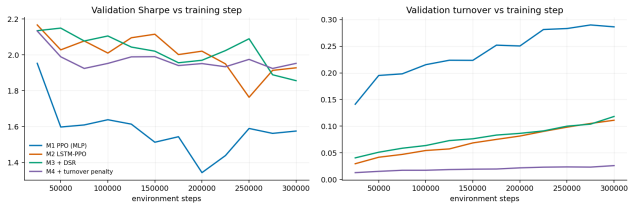


Figure 7.6: Validation Sharpe and turnover dynamics over 300,000 steps.

The grid sweep demonstrates that $\lambda = 0.1$ achieved the optimal trade-off on Fold 1 validation, maximizing the mean validation Sharpe ratio (4.40) while curtailing validation drawdown to −2.2%. At $\lambda = 0.01$, turnover remains high (0.040), whereas at $\lambda = 0.3$, the penalty stifles necessary reallocations. We explicitly note as a limitation that $\lambda_{\text{turn}} = 0.1$ was selected on Fold 1 alone across two seeds and applied frozen across all folds. Finally, the validation curves show clear over-training beyond 100,000 steps, highlighting the critical importance of checkpoint early stopping.

Chapter 8 — Limitations and Future Work

8.1 Methodological and Empirical Limitations

While this project provides a rigorous, controlled investigation into reinforcement learning for portfolio management, several realistic limitations must be acknowledged:

8.1.1 Passive Benchmark Outperformance and Survivorship Bias

The equal-weight 1/N strategy demonstrated superior cumulative profit and Sharpe ratio compared to all deep RL agents. This is attributable to two primary factors:

1. **Survivorship Bias in Equity Selection:** The 10 stocks selected (e.g., Reliance, TCS, HDFC Bank, ICICI Bank) represent established corporate leaders that survived and prospered over the 2011–2025 window. In an asset universe characterised by strong upward drift, a passive, fully-invested allocation captures market beta at negligible transaction costs (0.61%).
2. **Cash Drag from Learned Risk Aversion:** The continuous Dirichlet policy learned to maintain cash buffers (up to 32.4% in M4) to guard against volatility. While this successfully curbed

maximum drawdown to -16.04%, it penalized nominal capital compounding during bull markets.

8.1.2 Market Regime Dependence and Single-Epoch Hyperparameter Selection

As disclosed in Chapter 7, the turnover regularisation coefficient $\lambda_{\text{turn}} = 0.1$ was tuned strictly on Fold 1 validation across two seeds. Subsequent folds encompassed vastly different market regimes—notably the COVID-19 crash of March 2020 (Fold 2) and the global inflationary cycle of 2022 (Fold 3). A constant hyperparameter tuned in a calm market proved overly restrictive during volatile recovery phases where rapid rebalancing was necessary.

8.1.3 Simplified Execution and Market Impact Assumptions

The environment models execution costs as a fixed proportional fee $c_{\text{trans}} = 0.001$ (10 bps). While appropriate for retail and liquid institutional execution, this model neglects:

- Non-linear market impact and price slippage for large order sizes.
- Intraday execution dynamics (trades are executed at the official daily closing price).
- Short-selling constraints and margin financing costs.

8.1.4 Computational and Stochastic Architecture Constraints

Due to computational constraints, policies were restricted to a single-layer LSTM with 512 hidden units, trained for 300,000 steps per run. Furthermore, continuous Dirichlet sampling exhibits inherent stochasticity, leading to noticeable variance across random seeds (e.g., Fold 2 return std $\pm 14.8\%$ in M4).

8.2 Directions for Future Research

The insights gained from this study motivate several promising avenues for future inquiry:

1. **Dynamic and Adaptive Turnover Regularisation:** Rather than maintaining a static penalty λ_{turn}, future models could employ adaptive penalty scheduling conditioned on real-time market volatility (e.g., VIX) or learned meta-policy networks.
2. **Attention and Transformer Backbones:** Replacing recurrent LSTM cells with temporal multi-head attention mechanisms (such as Temporal Fusion Transformers) could enhance the extraction of long-range cross-asset correlations.
3. **Macroeconomic and Alternative Feature Integration:** Augmenting technical indicators with macroeconomic variables (interest rates, crude oil prices, USD/INR foreign exchange rates) and sentiment features from financial disclosures.
4. **Dynamic Cash Hurdles:** Incorporating an active risk-free interest rate yield on the cash reserve to mitigate the harsh cash drag observed in equity bull markets.

Chapter 9 — Conclusion

9.1 Summary of Research Findings

This B.Tech. project developed, implemented, and empirically evaluated a continuous Actor–Critic Deep Reinforcement Learning framework for daily portfolio allocation across ten major equities on the

National Stock Exchange of India. By formulating the policy over a multivariate Dirichlet distribution, the agent generated continuous target allocations that strictly preserved the portfolio simplex manifold. Through an extensive, strictly controlled four-stage ablation study (M1–M4) evaluated across four rolling walk-forward folds and three random seeds (2018–2025), this study isolated the individual contributions of temporal memory, risk-sensitive rewards, and turnover regularisation.

9.2 Answers to Research Questions

We conclude by providing direct, evidence-based answers to the four core research questions:

- **RQ1 (Temporal Architecture):** *Does an LSTM-based recurrent encoder improve out-of-sample risk-adjusted returns and policy stability compared to an MLP encoder?*
  Answer: Yes, unequivocally. Transitioning from the instantaneous MLP (M1) to the sequence-aware LSTM (M2) improved the out-of-sample Sharpe ratio by $+0.201$ (from 0.714 to 0.915, 95% CI $[+0.156, +0.246]$) with consistent improvements across all four folds (4/4). Moreover, the LSTM reduced daily turnover from 0.186 to 0.080 and cut total transaction costs by more than half (from 9.68% to 4.27%), demonstrating that temporal memory is essential for filtering market noise and preventing counterproductive daily portfolio churning.
- **RQ2 (Risk-Adjusted Reward):** *Does transitioning from raw net return to the Differential Sharpe Ratio (DSR) lead to improved Sharpe ratios and reduced drawdowns?*
  Answer: Partially, with nuanced trade-offs. The DSR reward (M3) yielded an incremental Sharpe delta of $+0.003$ ($[-0.295, +0.222]$) relative to M2. While DSR demonstrated strong performance in later regimes (Sharpe 1.328 in Fold 3 and 1.164 in Fold 4), its online variance estimation slightly depressed cumulative profit during aggressive bull markets (26.93% vs. 28.23%).
- **RQ3 (Turnover Regularisation):** *Does augmenting the reward with an explicit quadratic turnover penalty effectively suppress portfolio turnover and transaction costs without destroying wealth accumulation?*
  Answer: Yes, decisively. Incorporating the quadratic turnover penalty in Model M4 suppressed average daily turnover from 0.079 to 0.019 ($[-0.082, -0.044]$, improved in 12/12 pairs), reducing transaction costs to just 0.97% of starting capital. Crucially, M4 delivered the shallowest maximum drawdown of any evaluated model (-16.04%, an improvement of $+3.90\%$ over M3), successfully demonstrating that explicit turnover regularisation stabilizes portfolio execution under proportional transaction costs.
- **RQ4 (Benchmark Competitiveness):** *Can a model-free continuous actor-critic agent outperform a passively rebalanced equal-weight benchmark in a high-friction Indian equity universe?*
  Answer: No. The passive Equal-Weight benchmark achieved superior cumulative return (36.48%) and Sharpe ratio (1.131). Our gap decomposition revealed that this underperformance was predominantly driven by *cash drag* (-42.35% in M4) caused by the policy's learned risk-averse allocation to cash (32.4%) during an extended secular bull market. In asset universes characterised by strong macroeconomic upward drift, overcoming the beta advantage and negligible cost (0.61%) of passive 1/N diversification remains an formidable challenge for active reinforcement learning agents.

9.3 Concluding Remarks

In summary, this research demonstrates that deep reinforcement learning can autonomously discover disciplined risk-management and cost-conscious trading policies in real financial markets. While active outperformance against passive broad-market equities is bounded by cash drag and execution costs,

the controlled ablation methodology established in this thesis provides a rigorous scientific template for future investigations in quantitative financial machine learning.

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